

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer building full-stack features (Next.js + Firebase/GCP) end to end on a provider-facing clinical decision support product from its earliest version, in a fast-moving, ambiguous early-stage environment.
- Extending a **Firebase Cloud Functions** backend that builds medication alternatives from **RxNorm** normalization and PubMed/guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility, keeping the AI grounded and information-extraction-driven rather than a black box.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's **NLP** text-analysis platform (Python, **FastAPI**, sentence-transformers, React), productionizing a published research method (CCR) into a self-serve web tool while working directly with the lab's science team.
- Built the platform's backbone: a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and a structured data-quality warning system; grew the backend suite to **40 tests** (hermetic CI).

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage **SaaS** backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Diagnosed and resolved production issues across tenants, monitoring logs during releases and client integrations and shipping targeted fixes for tenant-specific incidents to keep the multi-tenant service stable.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic **information extraction** with LLM fallback and **cell-level provenance** for every value; hardened with **150+ tests**.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component of a KG-guided RAG pipeline (query-side decomposition, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a 4,905-question evaluation.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an AI research assistant whose agent fetches web sources and generates multi-step research digests, streaming to the client over WebSocket, with a durable workflow that survives process restarts and tool failures.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, TypeScript, JavaScript, Java, C++, SQL, Bash, MongoDB, PostgreSQL, SQLite

AI & NLP: LLM agents, RAG, information extraction, sentence-transformers, Hugging Face Transformers, PyTorch, evaluation

Retrieval / Search: RRF, cross-encoder reranking, MMR, multi-hop QA, cell-level provenance, Brave Search

Backend & Web: FastAPI, Node.js, Express, React, Next.js, REST APIs, async services, integration testing

Cloud & DevOps: Docker, Git, GitHub Actions, CI/CD, Cloudflare Workers, AWS, GCP, Azure

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with Stripe payments; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.